# Discrete Mathematics 1

**Midterm Examination**

Time allowed: 90 minutes  Closed book  No electronic devices

**Name:** _____ **Student ID:** _____

**Instructions:** Answer *all* questions. Show all working — correct answers without justification receive no credit. Clearly label each part.

**Score summary (examiner use only):**

| Q1 (4) | Q2 (4) | Q3 (4) | Q4 (4) | Total (16) |
|---|---|---|---|---|
| | | | | |

## Question 1 [4 points] — Counting

You have 9 distinct books: 4 on algebra, 3 on analysis, and 2 on topology.

(a) **[2 pts]** In how many ways can all 9 books be arranged on a shelf so that all algebra books form a contiguous block?

(b) **[2 pts]** In how many ways can 5 books be chosen so that at least 2 are algebra books and at least 1 is a topology book?

## Question 2 [4 points] — Combinatorial Identities & Partitions

(a) **[2 pts]** Prove the following identity by a combinatorial argument:

$$\sum_{k=0}^{n} k\binom{n}{k} = n\,2^{n-1}.$$

*Hint: think of choosing a committee together with a designated chair.*

(b) **[1 pt]** Evaluate the Bell number $B_4$.

(c) **[1 pt]** Evaluate the Stirling number $S(6,3)$.

## Question 3 [4 points] — Inclusion-Exclusion

(a) **[2 pts]** How many integers from 1 to 1200 are divisible by none of 3, 8, and 25?

(b) **[2 pts]** Ten people randomly pick up one of ten distinct hats. In how many ways do *exactly* 3 people get their own hat?

## Question 4 [4 points] — Generating Functions

(a) **[2 pts]** Find the generating function of the sequence $a_n = \binom{n+2}{2}$, $n = 0, 1, 2, \ldots$, as a closed-form rational function of $x$.

(b) **[2 pts]** Let $f(x) = \sum_{n=0}^{\infty} a_n x^n$ be the generating function of $(a_n)$. Find the generating function of

$$b_n = a_{n+1} - 3\,a_n, \qquad n = 0, 1, 2, \ldots,$$

expressed in terms of $f(x)$ and $a_0$.

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*Total: 16 points. Grade boundary: 9/16 pass, 13/16 distinction.*

# Discrete Mathematics 1

**Midterm Examination — Model Solutions**

## Question 1 — Counting

(a) Treat the 4 algebra books as a single block. We then have 6 objects to arrange (the block, 3 analysis books, 2 topology books), giving 6! orders. The 4 algebra books can be arranged within the block in 4! ways.

$$6! \times 4! = 720 \times 24 = \mathbf{17\,280}.$$

(b) We need to choose 5 books with at least 2 algebra (A) and at least 1 topology (T); the rest are analysis (An). Write $(a, t, \mathrm{an})$ for the counts with $a + t + \mathrm{an} = 5$, $a \in \{2, 3, 4\}$, $t \in \{1, 2\}$, $\mathrm{an} \geq 0$, and $\mathrm{an} \leq 3$.

| $a$ | $t$ | an | count |
|---|---|---|---|
| 2 | 1 | 2 | $\binom{4}{2}\binom{2}{1}\binom{3}{2} = 6 \cdot 2 \cdot 3 = 36$ |
| 2 | 2 | 1 | $\binom{4}{2}\binom{2}{2}\binom{3}{1} = 6 \cdot 1 \cdot 3 = 18$ |
| 3 | 1 | 1 | $\binom{4}{3}\binom{2}{1}\binom{3}{1} = 4 \cdot 2 \cdot 3 = 24$ |
| 3 | 2 | 0 | $\binom{4}{3}\binom{2}{2}\binom{3}{0} = 4 \cdot 1 \cdot 1 = 4$ |
| 4 | 1 | 0 | $\binom{4}{4}\binom{2}{1}\binom{3}{0} = 1 \cdot 2 \cdot 1 = 2$ |

Total: $36 + 18 + 24 + 4 + 2 = \mathbf{84}$.

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## Question 2 — Combinatorial Identities & Partitions

(a) **Combinatorial proof.** Both sides count the number of ways to form a non-empty committee from $n$ people and designate one member as chair.

*Right-hand side.* Choose the chair first: $n$ ways. Each of the remaining $n - 1$ people independently joins or does not join the committee: $2^{n-1}$ ways. Total: $n\,2^{n-1}$.

*Left-hand side.* First choose the committee size $k$ ($1 \leq k \leq n$), then choose the $k$ members ($\binom{n}{k}$ ways), then designate one of them as chair ($k$ ways). Summing over all sizes gives $\sum_{k=0}^{n} k\binom{n}{k}$. (The $k = 0$ term contributes zero.)

Since both sides count the same quantity the identity holds. $\square$

*Algebraic alternative.* Differentiate the binomial identity $(1 + x)^n = \sum_{k=0}^{n} \binom{n}{k} x^k$ with respect to $x$ to obtain $n(1 + x)^{n-1} = \sum_{k=0}^{n} k\binom{n}{k} x^{k-1}$. Setting $x = 1$ gives $\sum_{k=0}^{n} k\binom{n}{k} = n\,2^{n-1}$. $\square$

(b) Listing all set partitions of $\{1, 2, 3, 4\}$ by number of blocks:

| | |
|---|---|
| 1 block | $\{\{1,2,3,4\}\}$ (1 partition) |
| 2 blocks | $\{\{1\},\{2,3,4\}\}$, $\{\{2\},\{1,3,4\}\}$, $\{\{3\},\{1,2,4\}\}$, $\{\{4\},\{1,2,3\}\}$, $\{\{1,2\},\{3,4\}\}$, $\{\{1,3\},\{2,4\}\}$, $\{\{1,4\},\{2,3\}\}$ (7 partitions) |
| 3 blocks | $\{\{1,2\},\{3\},\{4\}\}$, $\{\{1,3\},\{2\},\{4\}\}$, $\{\{1,4\},\{2\},\{3\}\}$, $\{\{2,3\},\{1\},\{4\}\}$, $\{\{2,4\},\{1\},\{3\}\}$, $\{\{3,4\},\{1\},\{2\}\}$ (6 partitions) |
| 4 blocks | $\{\{1\},\{2\},\{3\},\{4\}\}$ (1 partition) |

Hence $B_4 = 1 + 7 + 6 + 1 = \mathbf{15}$.

(c) We use the recurrence $S(n,k) = k\,S(n-1,k) + S(n-1,k-1)$ with $S(n,1) = S(n,n) = 1$.

| $n\backslash k$ | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| 2 | 1 | 1 | | | |
| 3 | 1 | 3 | 1 | | |
| 4 | 1 | 7 | 6 | 1 | |
| 5 | 1 | 15 | 25 | 10 | 1 |
| 6 | 1 | 31 | **90** | 65 | 15 |

Key step: $S(6,3) = 3 \cdot S(5,3) + S(5,2) = 3 \cdot 25 + 15 = \mathbf{90}$.

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## Question 3 — Inclusion-Exclusion

(a) Let $A$, $B$, $C$ be the sets of integers in $[1, 1200]$ divisible by 3, 8, 25 respectively.

$$|A| = \left\lfloor \tfrac{1200}{3} \right\rfloor = 400, \quad |B| = \left\lfloor \tfrac{1200}{8} \right\rfloor = 150, \quad |C| = \left\lfloor \tfrac{1200}{25} \right\rfloor = 48.$$

Using $\operatorname{lcm}(3,8) = 24$, $\operatorname{lcm}(3,25) = 75$, $\operatorname{lcm}(8,25) = 200$, $\operatorname{lcm}(3,8,25) = 600$:

$$|A \cap B| = 50, \quad |A \cap C| = 16, \quad |B \cap C| = 6, \quad |A \cap B \cap C| = 2.$$

By inclusion-exclusion:

$$|A \cup B \cup C| = 400 + 150 + 48 - 50 - 16 - 6 + 2 = 528.$$

Numbers divisible by none of $3, 8, 25$: $1200 - 528 = \mathbf{672}$.

(b) Choose which 3 of the 10 people get their own hat: $\binom{10}{3} = 120$ ways. The remaining 7 people must all get a wrong hat, i.e. we need $D_7$, the number of derangements of 7 elements.

$$D_7 = 7! \sum_{k=0}^{7} \frac{(-1)^k}{k!} = 5040\big(1 - 1 + \tfrac{1}{2} - \tfrac{1}{6} + \tfrac{1}{24} - \tfrac{1}{120} + \tfrac{1}{720} - \tfrac{1}{5040}\big) = 1854.$$

Total number of ways: $\binom{10}{3} \cdot D_7 = 120 \times 1854 = \mathbf{222\,480}$.

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## Question 4 — Generating Functions

(a) Recall the standard identity $\displaystyle\sum_{n=0}^{\infty} \binom{n+r}{r} x^n = \frac{1}{(1-x)^{r+1}}$.

Setting $r = 2$:

$$\sum_{n=0}^{\infty} \binom{n+2}{2} x^n = \frac{1}{(1-x)^3}.$$

(b) The standard shift formula gives the generating function of $(a_{n+1})_{n\geq 0}$:

$$\sum_{n=0}^{\infty} a_{n+1}\, x^n = \frac{f(x) - a_0}{x}.$$

Therefore the generating function of $b_n = a_{n+1} - 3a_n$ is

$$g(x) = \frac{f(x) - a_0}{x} - 3f(x) = \frac{f(x) - a_0 - 3x\, f(x)}{x} = \frac{(1-3x)\, f(x) - a_0}{x}.$$

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*End of solutions.*